BOARD OF INTERMEDIATE EDUCATION, KARACHI

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

BIOLOGY

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

BOTANY

SECTION 'A' (Multiple Choice Questions) (20)

Note: (Options in the original are separated by star marks, shown here as A–D in printed order. Check marks on some options in the scan are a previous solver's marks and are not transcribed.)

Note: TIME: 1 Hour 30 Minutes — (40 Marks) — BOTANY 2021 (AS PER CONDENSED SYLLABUS)

Note: NOTE: i) This section consists of 20 part questions and all are to be answered. Each question carries ONE mark. ii) Do not copy the part questions in your answer book. Write only the answer in full against the proper number of the question and its part.

Q. SECTION 'A' (Multiple Choice Questions) (20)

Q. 1. Choose the correct answer for each from the given options:

Q1. Solution having low concentration as compared to the cell sap

(A) Hypertonic
(B) Hypotonic
(C) Isotonic
(D) Solute potential

Q2. In terms of number of chromosomes, the endosperm of send is:

(A) haploid
(B) Diploid
(C) Tripoid
(D) Quadreploid

Q3. Two polynucleotide chains of DNA are apart from each other by:

(A) 3A°A
(B) 4°A
(C) 20°A
(D) 34°A

Q4. A cell division without spindle is:

(A) Amitosis
(B) Mitosis
(C) Meiosis
(D) Apomines

Q5. This biome is called the "bread basket" of the world:

(A) Savanah
(B) Grass land
(C) Tundra
(D) Tropical Rain forest

Q6. This group of plants is found in soil of high salt concentration:

(A) Hydrophytes
(B) Mesophytes
(C) Halophytes
(D) Xerophytes

Q7. The breakdown of rocks due to environmental action is called:

(A) Topography
(B) Pedology
(C) Weathering
(D) Edaphic

Q8. The producer of a pond ecosystem is:

(A) Bacteria
(B) Phytoplankton
(C) Zooplankton
(D) Fungi

Q9. This condon is start condon:

(A) AUG
(B) UAG
(C) UAA
(D) UGA

Q10. Meristem responsible for secondary growth:

(A) Apical
(B) Intercalary
(C) Lateral
(D) Cork

Q11. During the process the translation, cells use this to make protein:

(A) RNA
(B) DNA
(C) Chromosomes
(D) Nucleus

Q12. The best soil for plant growth is:

(A) Sand
(B) Silt
(C) Loam
(D) Clay

Q13. Total aggregation of gene in a population is called:

(A) Chromosomes
(B) Multiple gene
(C) Gene pool
(D) Population

Q14. At High temperature plants produce these to preserve enzymes:

(A) Fructans
(B) Unsaturated fatty acids
(C) Heat shock proteins
(D) Phytoalexins

Q15. The genetic makeup of any individual is called:

(A) Allelomorph
(B) Phenotype
(C) Synapsis
(D) Genotype

Q16. During this stage of mitosis chromosomes arrange on equatorial plate:

(A) Prophase
(B) Metaphase
(C) Anaphase
(D) Teleophase

Q17. The two chromatids of the same chromosome are called:

(A) Chromonema
(B) Centromere
(C) Sister chromatid
(D) Kinetochore

Q18. Germination of seed inside the fruit is called:

(A) Epigeal
(B) hypogeal
(C) viviparous
(D) hypocotyl

Q19. Movement of plant unrelated to the direction of stimuli is known as:

(A) Nuatation
(B) Tropic
(C) Nastic
(D) Turgor

Q20. These tissues are commonly known as conducting tissue:

(A) Vascular tissues
(B) Cork Cambium
(C) Meristem
(D) Ground tissues

SECTION 'B' + SECTION 'C' (Subjective)

Q. SECTION 'B' (Short-Answer Questions) (12)

> Note: NOTE: 2. Attempt any four part questions. Each question carries ONE mark. Answer should not be exceed more than two lines

> Note: 3. Attempt any four part questions. Each question carries two marks. Answer should not exceed five lines.

Q. SECTION "C" (Detailed Answer Questions) (08)

> Note: NOTE: Attempt Two questions from this section. All questions carries equal marks.

> Note: نوٹ: سال 2020 میں گیارہویں اور بارہویں جماعت کے امتحانات نہیں ہوئے تھے۔

> نوٹ: سال 2020 میں گیارہویں اور بارہویں جماعت کے امتحانات نہیں ہوئے تھے۔

ZOOLOGY

SECTION 'A' (Multiple Choice Questions) (22)

> Note: (Options in the original are separated by star marks, shown here as A–D in printed order. Check marks on some options in the scan are a previous solver's marks and are not transcribed.)

> Note: TIME: 1 Hour 30 Minutes — (45 Marks) — ZOOLOGY 2021 (AS PER CONDENSED SYLLABUS)

> Note: NOTE: i) This section consists of 22 part questions and all are to be answered. Each question carries ONE mark. ii) Do not copy the part questions in your answer book. Write only the answer in full against the proper number of the question and its part.

Q. SECTION 'A' (Multiple Choice Questions) (22)

Q. 1. Choose the correct answer for each from the given options:

Qi. The process of removal of metabolic wastes is called:

(A) Homeostatis
(B) Excretion
(C) Osmoregulation
(D) Thermoregulation

Qii. Skeleton is totally absent in:

(A) Sponge
(B) Star fish
(C) Amoeba
(D) fish

Qiii. Kidney stone or calculi are 70% composed of:

(A) Calcium phosphate
(B) Calcium oxalate
(C) Magnesium phosphate
(D) Sodium oxalate

Qiv. It is not endocrine gland:

(A) Liver
(B) Pancreas
(C) Spleen
(D) Ovary

Qv. A clone exactly resemble with:

(A) Mother
(B) Father
(C) Diploid nucleus donor
(D) Haploid nucleus donor

Qvi. Lemark's theory based on following points except:

(A) Effect of environment
(B) Natural selection
(C) Use and discuss of organs
(D) Inheritance of acquired character

Qvii. The egg of hen is laid at the stage:

(A) Cleavage
(B) Blastula
(C) Morula
(D) Gastrula

Qviii. The concentration of oxygen in atmosphere is about:

(A) 21%
(B) 30%
(C) 50%
(D) 60%

Qix. The non-coding sequence of gene are:

(A) Codon
(B) Exon
(C) Intron
(D) Anti-codon

Qx. About 180 litres fluid is filtered by the kidney everyday, 1.2 litres is excreted in urine while remaining 178.8 litres.

(A) Lost as sweat
(B) stored in bladder
(C) reabsorbed into the blood
(D) becomes tissue fluid

Qxi. Each myosin filament is surrounded by:

(A) 4 actin filaments
(B) 6 actin filaments
(C) 8 actin filaments
(D) 12 actin filaments

Qxii. Goitre is caused by the lack of:

(A) iodine
(B) thyroxine
(C) protein
(D) lipids

Qxiii. The early division of fertilized egg is called:

(A) blastulation
(B) cleavage
(C) gastrulation
(D) neurulation

Qxiv. These are spindle shaped uninucleated muscles:

(A) smooth muscle
(B) striated and branched muscle
(C) striated and unbranched muscle
(D) elongated and binucleated muscle

Qxv. Hermaphodite is the term used for:

(A) Unisexual animals
(B) Bisexual animals
(C) Dioexious animals
(D) Multisexual animals

Qxvi. It is controlled by medulla oblongata in the brain:

(A) vision
(B) memory
(C) balance
(D) breathing

Qxvii. The enzyme used to seal the DNA is:

(A) Restriction enzyme
(B) Ligase enzyme
(C) Polymerase
(D) Endonuclease

Qxviii. Multifactorial defects include:

(A) one gene
(B) many genes
(C) many genes and environmental factors
(D) chromosomal abnormalities

Qxix. This bone is not present in the lipids

(A) Tibis
(B) Humerus
(C) Radius
(D) Ulta

Qxx. This gas is responsible for global warming

(A) Oxygen
(B) Nitrogen
(C) Carbon Dioxide
(D) ozone

Qxxi. ADH and oxytocin are secreted by:

(A) Anterior pituitary
(B) posterior pituitary
(C) pined gland
(D) Hypothalamus

Qxxii. Ovulation is initiated by this hormone:

(A) FSH
(B) LTH
(C) LH
(D) STH

SECTION 'B' + SECTION 'C' (Subjective)

Q. SECTION 'B' (Short-Answer Questions) (14)

> Note: NOTE: Attempt a SEVEN part questions in all. Select Four questions from Reasoning Questions and Three from Non-Reasoning Questions. All questions carry equal marks.

Q. SECTION "C" (Detailed Answer Questions) (09)

> Note: NOTE: Attempt any One questions from this section. All questions curry equal marks.

> Note: نوٹ: سال 2020 میں گیارہویں اور بارہویں جماعت کے امتحانات نہیں ہوئے تھے۔

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